

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- P80 assembly lubricant, Part Number 4918318
- Alco-Metalube® or Mercury® Quicksilver® 2-4-C Marine Grease.

Additional Service Items

- No additional service items required.

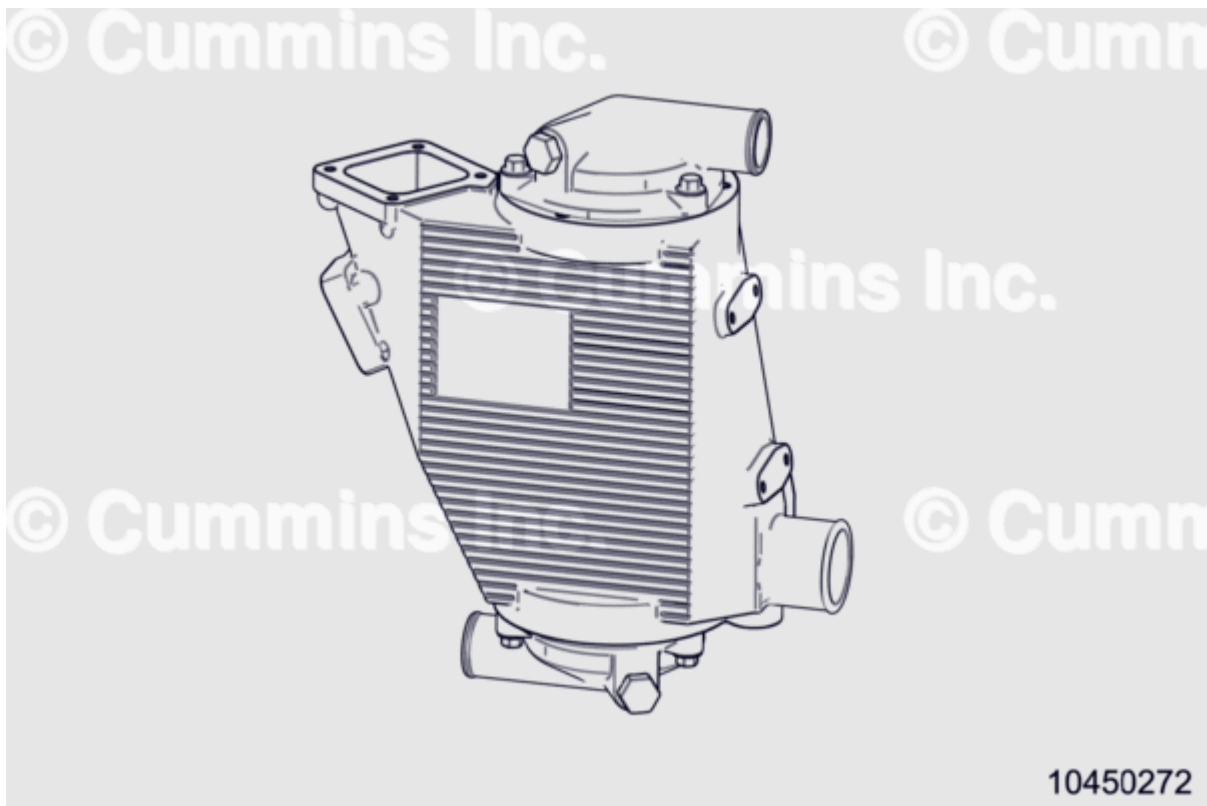
General Information

The aftercooler assembly is a vertical setup with the water inlet and outlet connections on the bottom and top respectively.

The aftercooler water inlet and outlet connections are made of bronze material.

Illustrations may **not** reflect the exact appearance of described parts.

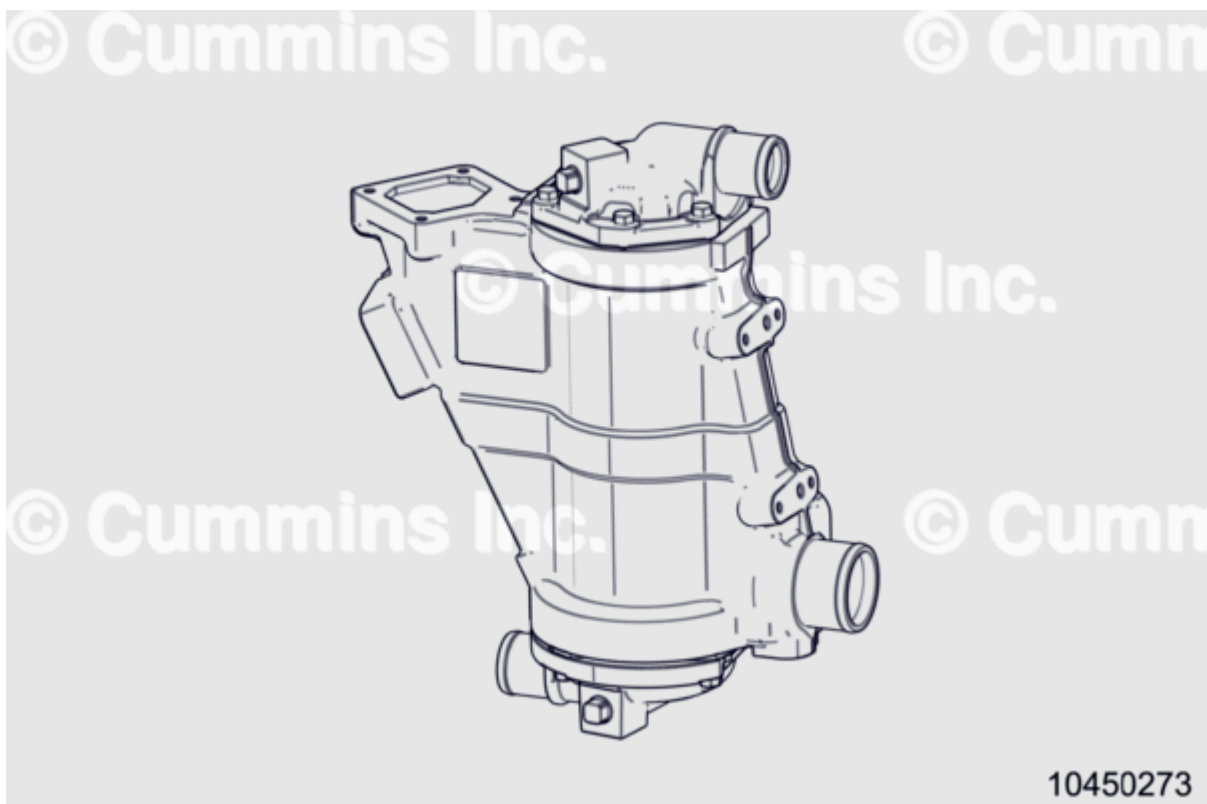
There are two component options for this application:



Option 1

Features:

- 3-bolt bronze end caps
- Continuous horizontal ribs demarking housing



Option 2

Features:

- 6-bolt bronze outlet cap (top)
- 4-bolt Bronze inlet cap (bottom)
- Single horizontal rib across housing

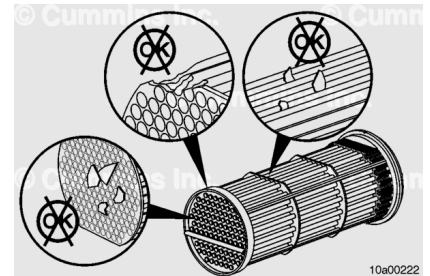
Initial Check

A turbocharger or sea water pump malfunction can cause damage to the aftercooler assembly.

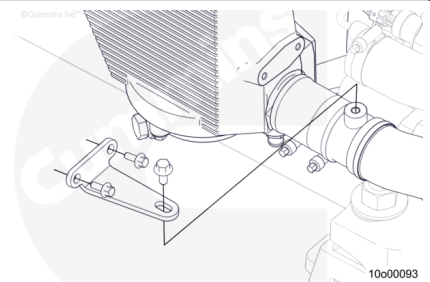
If a turbocharger malfunction or a sea water pump malfunction has occurred, remove the aftercooler element and inspect the element and housing for damage.

Inspect the zinc anodes to make sure they are still in one piece and have **not** broken.

See the Clean and Inspect step within this procedure.

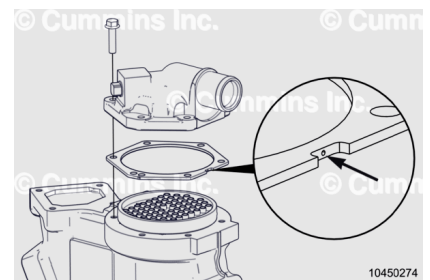


If necessary for repair, remove the bracket from the aftercooler.



Option 2 **only**:

While one end of aftercooler cores is fixed with mounting plate integrated into core tube header, the other end of the core can float. In support of this core float, a mounting spacer is sandwiched between end caps (opposite fixed core end) and aftercooler housing with o-ring on each side. This mounting spacer possesses a 1.5 - 2 mm [1/16 in – 5/64 in] weep hole through its wall to identify an improperly installed and/or defective o-ring. In the event of an issue with o-ring on water side of spacer, a water leak will be observed from weep hole with severity of leak dependent on extent of o-ring defect. When inspecting aftercooler assemblies, the weep hole should be checked to be free of any debris. As with cores and end caps, the bolt pattern in the mounting spacer controls consistent location of the weep hole.



Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

CAUTION

Use caution when disconnecting or removing fuel lines, replacing filters and priming the fuel system that fuel is not spilled or drained into the bilge area. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be discarded in accordance with local environmental regulations.

CAUTION

Zinc plugs expand and can break off during removal. Inspect the zinc plug to make sure it is in one piece. If not, it must be replaced with a new zinc plug, and the broken pieces must be retrieved from the aftercooler to prevent damage to components downstream in the sea water system.

- Close the sea water supply and return line(s). See equipment manufacturer service information.
- Close the fuel supply and return lines. See equipment manufacturer service information.
- Disconnect the battery power from the engine. See equipment manufacturer service information.
- Drain the sea water by removing the zinc plug at the bottom of the aftercooler assembly. Refer to Procedure 008-059 in Section 8. (</qs3/pubsys2/xml/en/procedures/40/40-008-059.html>)
- Remove the following, if equipped. Refer to Procedure 006-062 in Section 6. (</qs3/pubsys2/xml/en/procedures/40/40-006-062.html>)
 - Fuel cooler lower molded sea water hose to the aftercooler
 - Fuel cooler clip and mounting capscrew
 - Fuel drain flex hoses at the fuel cooler.

- Loosen the aftercooler top and bottom air connection hose clamps. Refer to Procedure 010-019 in Section 10. (</qs3/pubsys2/xml/en/procedures/40/40-010-019-tr.html>)
- Remove the aftercooler air box drain plug and drain the condensation from the aftercooler housing core.
- Disconnect the aftercooler air inlet connection hose. Refer to Procedure 007-011 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-011-tr.html>)
- Loosen the hose clamp on the water transfer connection and remove the aftercooler sea water discharge hose.
- Loosen the hose clamps connecting the aftercooler to the marine gear oil cooler. Refer to Procedure 008-041 in Section 8. (</qs3/pubsys2/xml/en/procedures/40/40-008-041.html>)
- Detach the oil dipstick tube if connected to the aftercooler. Refer to Procedure 007-011 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-011-tr.html>)

Remove

Remove the intake air heater cables, if equipped.

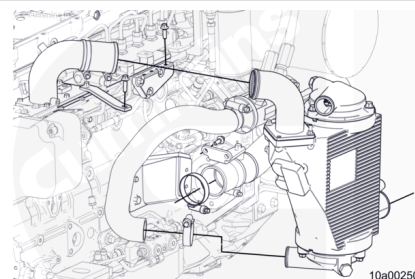
Remove the water hoses.

Remove air crossover hoses.

Remove the two capscrews that attach the front aftercooler bracket to the cylinder head.

Remove the two capscrews that attach the bottom aftercooler bracket to the flywheel housing.

Remove the aftercooler assembly.



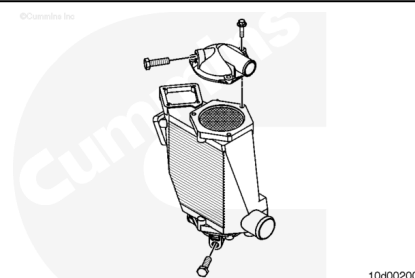
Disassemble

Option 1:

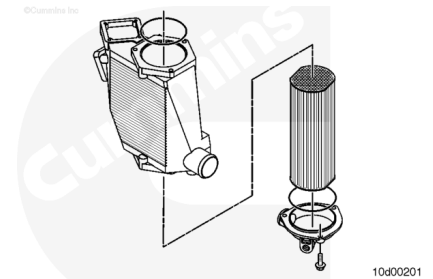
Mark or note the aftercooler inlet and outlet water connection locations.

Remove the zinc plugs from the upper water transfer connection.

Remove the three capscrews and upper water connection.



The aftercooler core assembly will slide out of the cooler housing when removing the lower aftercooler water connection. To reduce the possibility of damaging the core, lay the aftercooler assembly on its side to remove the connection and the core.



Lay the aftercooler assembly on its side.

The core assembly has two flat sides which **must** be oriented toward the aftercooler air intake and outlet sides of the aftercooler.

Remove the lower connection cover and the core.

If unable to remove the core by hand, a press should be used. A load spreading plate should be used to press on the outer ring of the core only and not on the ends of the tubes. The plate should have an outer diameter of 120 mm [4.72 in] and a step to fit inside the aftercooler endplate lip of 116 mm [4.57 in]. It can take up to 1000 lbs to press the core out.

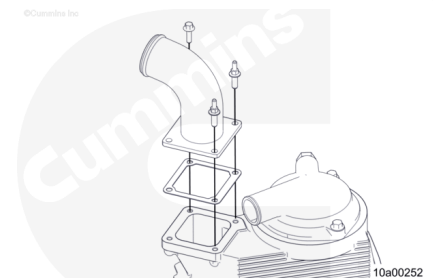
If the core is removed with a hammer (or impact), it can **not** be reused.

Remove the upper and lower sealing o-rings.

Remove the aftercooler check valve and zinc anodes.

If necessary for repair, remove the intake elbow and gasket from the aftercooler.

Document the orientation of the intake elbow and the location of the studded capscrews for the oil dipstick tube support bracket prior to removal.



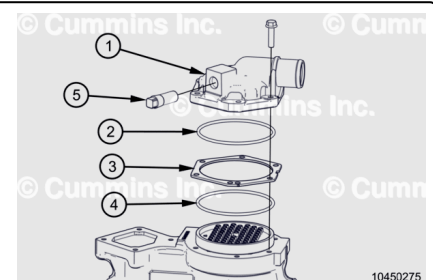
Option 2:

Mark or note the aftercooler front water connection, mounting spacer sealing ring, and rear cover orientation.

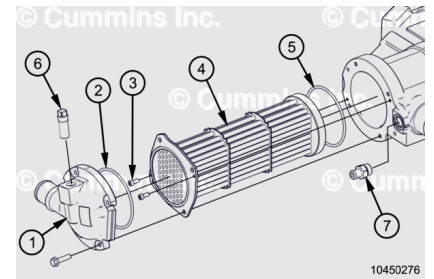
Remove the capscrews and rear cover connection (1).

Remove the rear o-ring (2), mounting spacer sealing ring (3), and second rear o-ring (4).

Remove the zinc plug (5).



The aftercooler core assembly will slide out of the cooler housing when removing the front aftercooler water connection cover. To reduce the possibility of damaging the core, lay the aftercooler assembly on its side to remove the connection cover and the core.



The core assembly has two flat sides which **must** be oriented toward the aftercooler air intake and outlet sides of the aftercooler.

Remove the aftercooler check valve (7) and zinc anode (6).

Remove the capscrews and front water connection cover (1).

Remove the front sealing o-ring (2).

Remove the core position fasteners (3)

Remove the core (4).

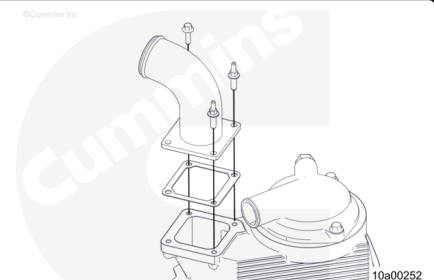
Remove the front sealing o-ring (5) around the core outside diameter.

If unable to remove the core by hand, a press should be used. A load spreading plate should be used to press on the core. The plate should have an outer diameter slightly less than that of the core.

If the core is removed with a hammer, or impact, it can **not** be reused.

If necessary for repair, remove the intake elbow and gasket from the aftercooler.

Document the orientation of the intake elbow and the location of the studded capscrews for the oil dipstick tube support bracket prior to removal.



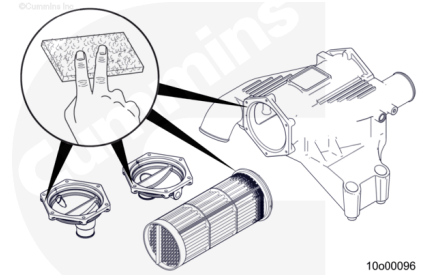
Clean and Inspect for Reuse

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Seawater leaks in the aftercooler can cause severe engine damage. Inspection for reuse and use of Cummins® seal kits are the extremely important.



Note : The aftercooler fins can be cleaned by ultrasonic cleaning, power washing, or steam cleaning. Use care to **not** damage the fins. The aftercooler core tubes can be cleaned with acidic marine cleaners such as hull cleaners, muriatic acid, or by rodding with a brass or aluminum rod.

Clean the aftercooler core o-ring sealing areas with a Scotch-Brite™ pad or sandpaper. Do not use power tools to clean the housing (removing material will result in aftercooler leaking).

Clean the aftercooler housing sealing areas and internal passages with Scotch-Brite™ pad or sandpaper. Do **not** use power tools to clean the housing. Removing seal area material will result in aftercooler leaking.

Clean water inlet and outlet connection sealing areas with a Scotch-Brite™ pad or sandpaper. Do **not** use power tools (removing seal area material will result in aftercooler leaking). White anode deposits in the water connections can be removed with acidic marine cleaners such as hull cleaners or muriatic acid.

Dry with compressed air.

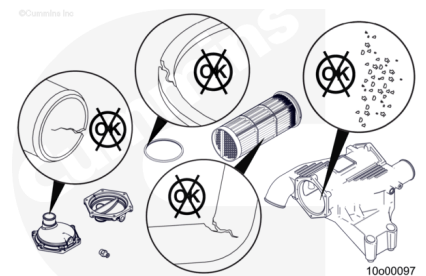
Inspect the housing for signs of seawater leaking into the intake air. Large amounts of corrosion will be present in the middle of the housing if leaking occurs. If there is a leak and the source can **not** be found, the entire aftercooler **must** be replaced.

Inspect the aftercooler assembly housing and water inlet and outlet connections for cracks or other damage.

Inspect the aftercooler housing, sealing ring, and core for the following:

- Deterioration of sealing surfaces
- Pitting/scratches over 0.25 mm [0.010 in] deep.

Inspect the molded hoses for damage. Replace if damage is found.



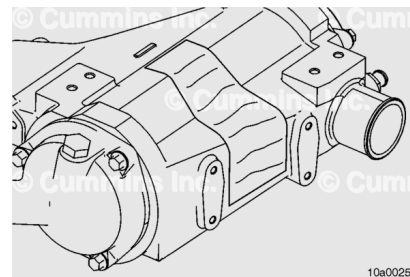
Inspect the heat shielding installed on the aftercooler.

If the shield is torn or sufficiently worn such that the black molded hose is exposed in the area to be shielded, the shield should be replaced.

Note : If an issue is found with the heat shield on the aftercooler, the gear oil cooler sea water hose should be inspected for damage. Refer to Procedure 008-041 in Section 8. (</qs3/pubsys2/xml/en/procedures/40/40-008-041.html>)

To install new heat shield on the aftercooler, peel off backing to expose adhesive, align between bosses and edge of housing, and press on.

Note : If the aftercooler is replaced, a new heat shield should also be installed.

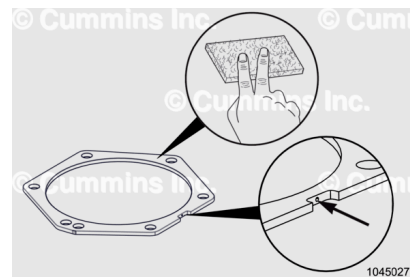


Option 2 **only**:

Inspect and clean the mounting spacer sealing ring weep hole.

Clean sealing areas with a Scotch-Brite™ pad or sandpaper. Do **not** use power tools (removing seal area material will result in aftercooler leaking).

Dry with compressed air.

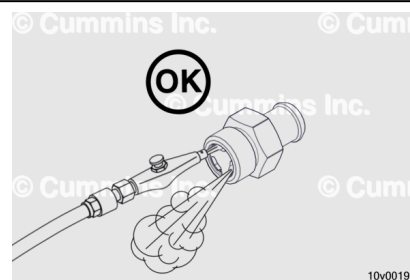


Inspect the aftercooler check valve / condensation drain for the following conditions:

- The internal piston should rattle when shaken.
- Valve should flow air freely under low pressure.
- Valve should close under high pressure. A pressure of over 0.5 psi from the threaded side of the valve should close the valve.

Operation can be checked using compressed air. Replace the valve if **not** operating properly.

Inspect the valve for excessive deposits on its interior surface. White deposits 1/16" thick or a clogged valve indicate an internal seawater leak requiring the aftercooler to be disassembled and inspected.



Assemble

Option 1:

Assemble the aftercooler in a vertical position.

Coat the following with Alco-Metalube® or Mercury® Quicksilver® 2-4-C Marine Grease.

- Outer diameter of the core endplates.
- Machined surfaces at the ends of the aftercooler housing.
- Machined surfaces on the inside of the water inlet and outlet connections.

Install the aftercooler core into the aftercooler housing. Align the marks so one flat side of the core is facing the air inlet side of the housing.

Apply lubricant from the seal kit to o-rings before they are installed.

Install a new o-ring around the lower core outside diameter.

Install the water inlet and outlet connection. Tighten the capscrews.

Torque Value: 43 n•m [32 ft-lb]

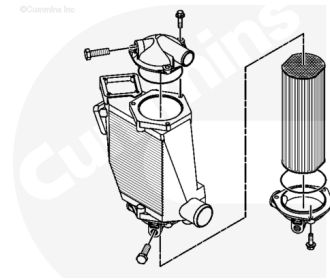
Install a new o-ring on the aftercooler check valve, if needed.

Install the aftercooler check valve.

Torque Value: 30 n•m [22 ft-lb]

Inspect and install the zinc plugs. Tighten. Refer to Procedure 008-059 in Section 8.

(/qs3/pubsys2/xml/en/procedures/40/40-008-059.html)



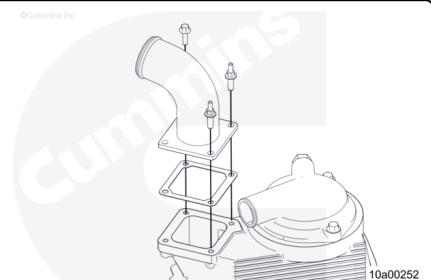
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Install the intake elbow, gasket, and mounting capscrews to the aftercooler.

Note : A new gasket should be used on the intake elbow connection.

Note : The studded capscrews for the oil dipstick tube support bracket should be installed closest to the aftercooler water inlet and outlet connection.

Torque Value: 24 n•m [212 in-lb]



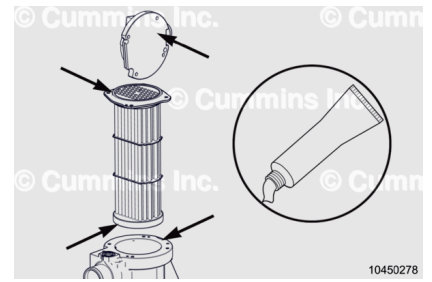
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Option 2:

Assemble the aftercooler in a vertical position.

Coat the following with Alco-Metalube® or Mercury® Quicksilver® 2-4-C Marine Grease.

- Outer diameter of the core endplates.
- The flat machined surfaces and the first inch of the bore in the housing at both ends of the aftercooler housing.
- Machined surfaces on the inside of the front and rear cover.

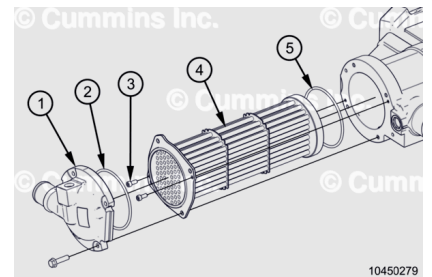


Apply Alco-Metalube® or Mercury® Quicksilver® 2-4-C Marine Grease to o-rings before they are installed.

Install sealing o-ring (5) onto the core (4) and slide back all the way to the ridge on the core.

Install the aftercooler core (4) into the aftercooler housing.

Install the core positioning fasteners (3).



Torque Value: 8 n•m [71 in-lb]

Install a new o-ring (2) around the front core outside diameter.

The core assembly has two flat sides which **must** be oriented toward the aftercooler air intake and outlet sides of the aftercooler.

Install the front water connection cover (1) and tighten the capscrews in a crisscross pattern.

The front cover should be orientated as noted during disassembly.

Torque Value:

Bronze water connection cover 24 n•m [212 in-lb]

Install a new o-ring (4) around the rear core outside diameter.

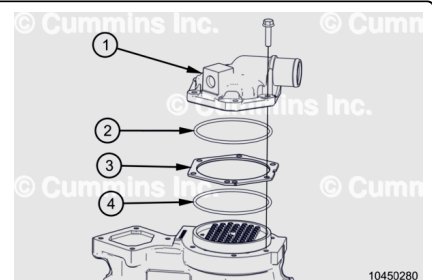
Coat the entire sealing ring (3) with Alco-Metalube® or Mercury® Quicksilver® 2-4-C Marine Grease.

Install the mounting spacer sealing ring (3) on the rear of the core.

Note: The mounting spacer sealing ring should be installed with the weep hole orientated as noted during disassembly.

Install the new front o-ring (2) around the rear core outside diameter.

Install the rear cover (1) and tighten the capscrews in a crisscross pattern.



Note: The rear cover should be orientated as noted during disassembly.

Torque Value:

Bronze connection cover 24 n•m [212 in-lb]

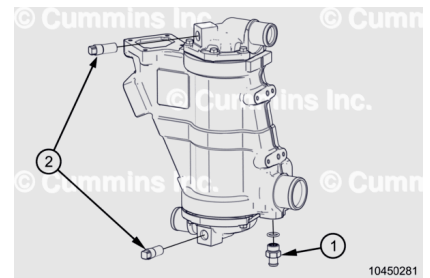
If removed, install the check valve and zinc anodes.

Install the new o-ring on the aftercooler check valve (1), if needed. Lubricate the o-ring with P80 assembly lubricant, Part Number 4918318, to aid in installation.

Install the aftercooler check valve.

Torque Value: 30 n•m [22 ft-lb]

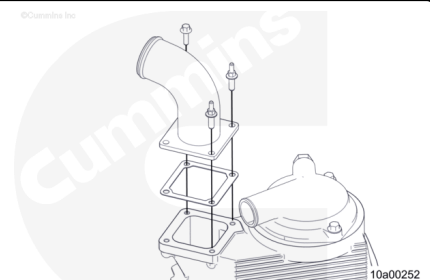
Install the zinc plugs and tighten. Refer to Procedure 008-059 in Section 8. (</qs3/pubsys2/xml/en/procedures/40/40-008-059.html>)



Install the intake elbow, gasket, and mounting capscrews to the aftercooler.

A new gasket should be used on the intake elbow connection.

The studded capscrews for the oil dipstick tube support bracket should be installed closest to the aftercooler water inlet and outlet connection.



Torque Value: 24 n•m [212 in-lb]

Pressure Test

⚠ WARNING ⚠

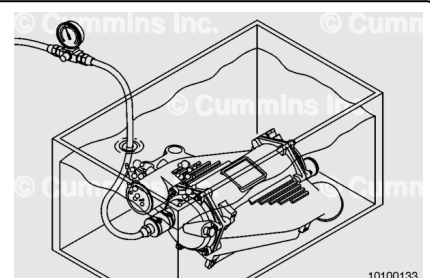
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Pressure test the sea water side of the aftercooler assembly.

Apply 276 kPa [40 psi] air pressure.

Lower the assembly into a tank of water and check for leaks.

Completely dry the aftercooler with compressed air.



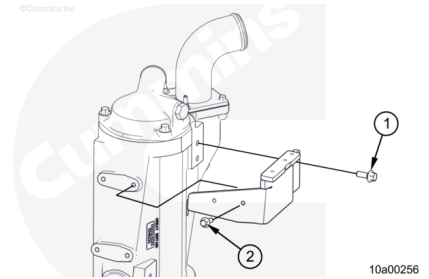
Install

If removed, install the top mounting bracket and two M10 capscrews onto the aftercooler.

Torque Value: 46 N·m [34 ft-lb]

Install the aftercooler bottom bracket and two M10 capscrews onto the aftercooler.

Torque Value: 35 N·m [25 ft-lb]



Note : Leave the aftercooler mounting capscrews loose to allow the aftercooler to move during installation.

Use a hoist to lift the aftercooler into position on the engine.

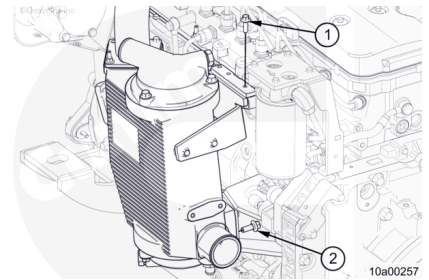
Install the two M12 capscrews to attach the top aftercooler bracket to the cylinder head. Do **not** tighten the capscrews.

The lifting bracket and circuit breaker bracket **must** be on top of the aftercooler bracket.

Install two M12 capscrews to attach the bottom aftercooler bracket to the flywheel housing.

Tighten the aftercooler mounting capscrews.

Torque Value: 80 N·m [59 ft-lb]



Install the air crossover tube and hoses. Tighten the hose clamps.

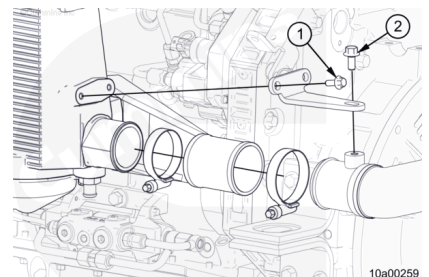
Torque Value: 8 N·m [71 in-lb]

Install the water hoses. Tighten the hose clamps.

Torque Value: 8 N·m [71 in-lb]

Install the intake air heater cables, if equipped.

Torque Value: 8 N·m [71 in-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

⚠ CAUTION ⚠

Use caution when disconnecting or removing fuel lines, replacing filters and priming the fuel system that fuel is not spilled or drained into the bilge area. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be discarded in accordance with local environmental regulations.

⚠ CAUTION ⚠

Zinc plugs expand and can break off during removal. Inspect the zinc plug to make sure it is in one piece. If not, it must be replaced with a new zinc plug, and the broken pieces must be retrieved from the aftercooler to prevent damage to components downstream in the sea water system.

- Attach the oil dipstick tube, if connected, to the aftercooler. Refer to Procedure 007-011 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-011-tr.html>)
- Install the following, if removed. Refer to Procedure 006-062 in Section 6. (</qs3/pubsys2/xml/en/procedures/40/40-006-062.html>)
 - Fuel cooler lower molded sea water hose to the aftercooler
 - Fuel cooler clip and mounting capscrew
 - Fuel drain flex hoses at the fuel cooler.
- Connect the aftercooler air inlet connection to the turbocharger. Refer to Procedure 010-019 in Section 10. (</qs3/pubsys2/xml/en/procedures/40/40-010-019-tr.html>)
- Install the aftercooler sea water discharge to the marine gear cooler hose. Refer to Procedure 008-041 in Section 8. (</qs3/pubsys2/xml/en/procedures/40/40-008-041.html>)
- Tighten the aftercooler top and bottom air connection hose clamps.
- Install the dipstick tube brace. Refer to Procedure 007-011 in Section 11. (</qs3/pubsys2/xml/en/procedures/40/40-007-011-tr.html>)
- Connect the battery power to the engine. See equipment manufacturer service information.
- Open the sea water supply and return line(s). See equipment manufacturer service information.
- Open the fuel supply and return lines. See equipment manufacturer service information.
- Operate the engine. Check for leaks.

